

Falcon 9 project

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Executive summary

- **Key Findings:**
- **Operational Hubs:** Florida serves as the primary operational base for SpaceX. Among all locations, KSC LC-39A recorded the highest proportional success rate (40% of all successful missions).
- **Payload & Orbit Dynamics:** Contrary to expectations, higher payload masses often correlated with higher success rates. This was heavily driven by high-volume missions to VLEO (e.g., Starlink), which achieved near-perfect success despite carrying maximum payloads.
- **Historical Progress:** Early booster versions experienced higher failure rates at lower payloads, reflecting developmental testing. The first successful ground-pad landing on 22 December 2015 marked a pivotal shift in reliability.
- **Predictive Modeling:** After evaluating multiple algorithms, **Logistic Regression** emerged as the most reliable predictive model, offering the best balance of accuracy (0.85) and R2 score (0.83). Support Vector Machines (SVMs) proved unviable due to excessive computational time.

Introduction

- **Project Overview:** This project explores the historical launch data of SpaceX's Falcon 9 rocket to identify the key variables that influence first-stage landing success.
- **Objectives:**
 - To conduct Exploratory Data Analysis (EDA) using SQL and Data Visualisation tools to uncover trends related to launch sites, payload masses, and orbit types.
 - To evaluate historical booster performance and landing outcomes across different operational periods.
 - To develop and compare Machine Learning models to accurately predict future landing outcomes.

Data Collection -API

- Data extraction: Requested .json file and parsed the SpaceX launch data using the GET request
- Feature extraction: Defined the function to decipher Booster version, Launch Site, Payload data, and other core data, such as landing outcome and number of flights
- Scope & Filtering: Filtered the dataframe to only include Falcon 9 launches
- Data cleaning: Imputed missing values in Payload mass column with the mean value

URL: <https://github.com/jingsann/Falcon-9-project/blob/4a37e516252e814ca6d9be837e05791acfe55423/jupyter-labs-spacex-data-collection-api.ipynb>

Data Collection – Web Scrapping

- Data extraction: Requested the Falcon 9 Launch Wiki page from its URL and parsed HTML using BeautifulSoup
- Feature extraction: Extracted all column/variable names from the HTML table header <th> using `extract_column_from_header()`
- Data conversion: Used defined helper functions to extract the data and created a dataframe by parsing the launch HTML tables
- Collected variables: Obtained Flight No., Launch Site, Payload Name and Mass, Orbit, Launch Outcome, Booster Landing Status, Date & Time

URL: <https://github.com/jingsann/Falcon-9-project/blob/4a37e516252e814ca6d9be837e05791acfe55423/jupyter-labs-webscraping.ipynb>

Data Wrangling

Exploring and Analysing the Dataset

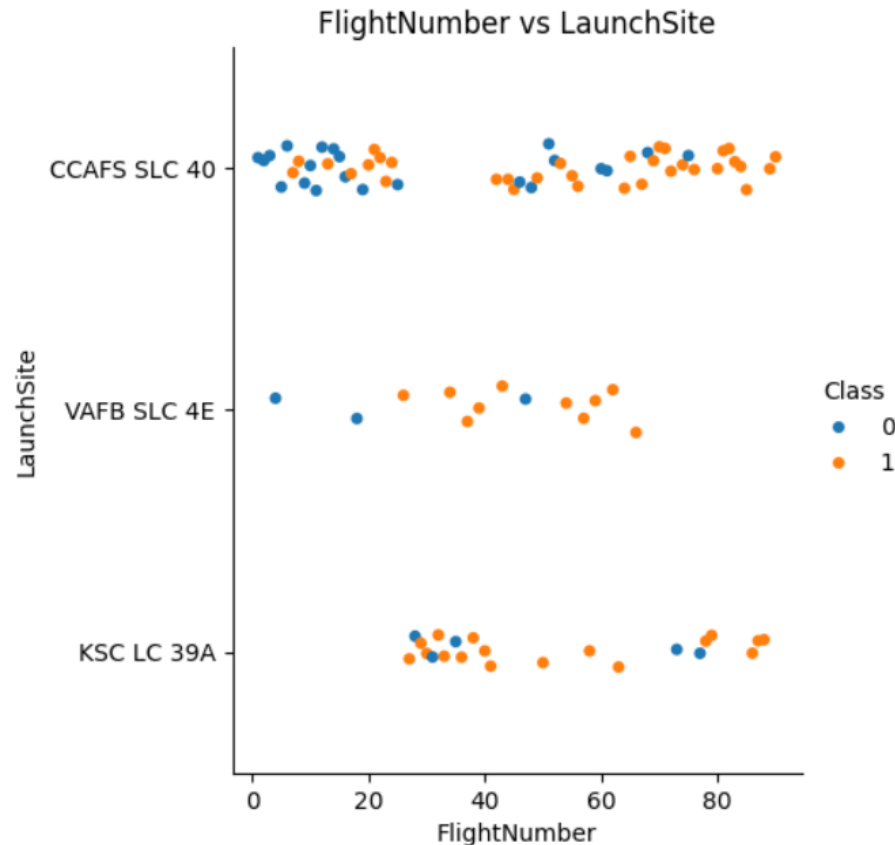
- Calculated the number of launches on each site using `.value_counts()`
- Calculated the number and occurrence of each orbit, and filtered GTO orbit using `.value_counts()` and nested conditions
- Analysed the number and occurrence of mission outcome of the orbits, and categorised into good and bad outcomes

Create Landing Outcome Labels

- Assigned value 0 for bad outcomes and 1 for good outcomes
- Stored the value in the new column called “Class”

- URL: <https://github.com/jingsann/Falcon-9-project/blob/11cecb71e163b202138ce0a63d006fae62c105ab/labs-jupyter-spacex-Data%20wrangling.ipynb>

EDA with Data Visualisation



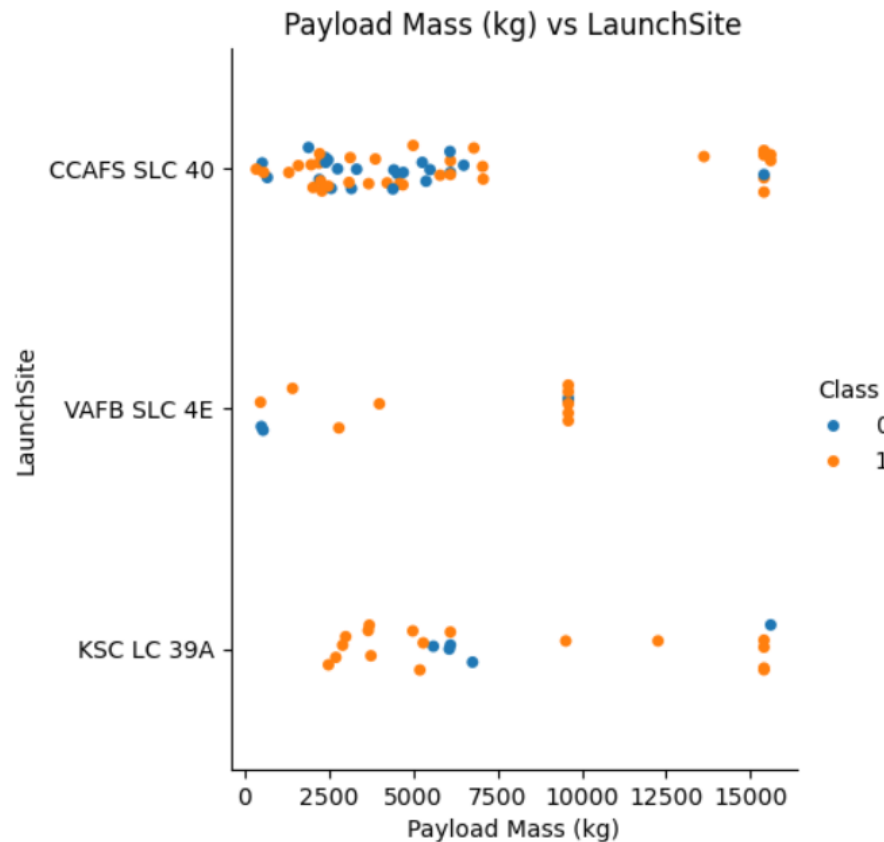
Key Insights:

- **Overall Trend:** The higher FlightNumber tended to have a greater success rate for every launch site.
- **CCAFS SLC 40:** It performed poorly with approximately half of the attempts ended up failing. However, after FlightNumber 40, the performance was exceptional with less than 10 flights in class 0.
- **VAFB SLC 4E & KSC LC 39A:** The overall performance for both landing sites was stellar with only 3 failed attempts in total for VAFB SLC 4E, and 5 failures for KSC LC 39A, respectively.

Figure 1: Flight Number vs Launch Site landing outcome distribution.

- URL: <https://github.com/jingsann/Falcon-9-project/blob/8db94669c76026d1c6d05efbe03acf18b2aa1886/Spacex%20data%20preparation.ipynb>

EDA with Data Visualisation



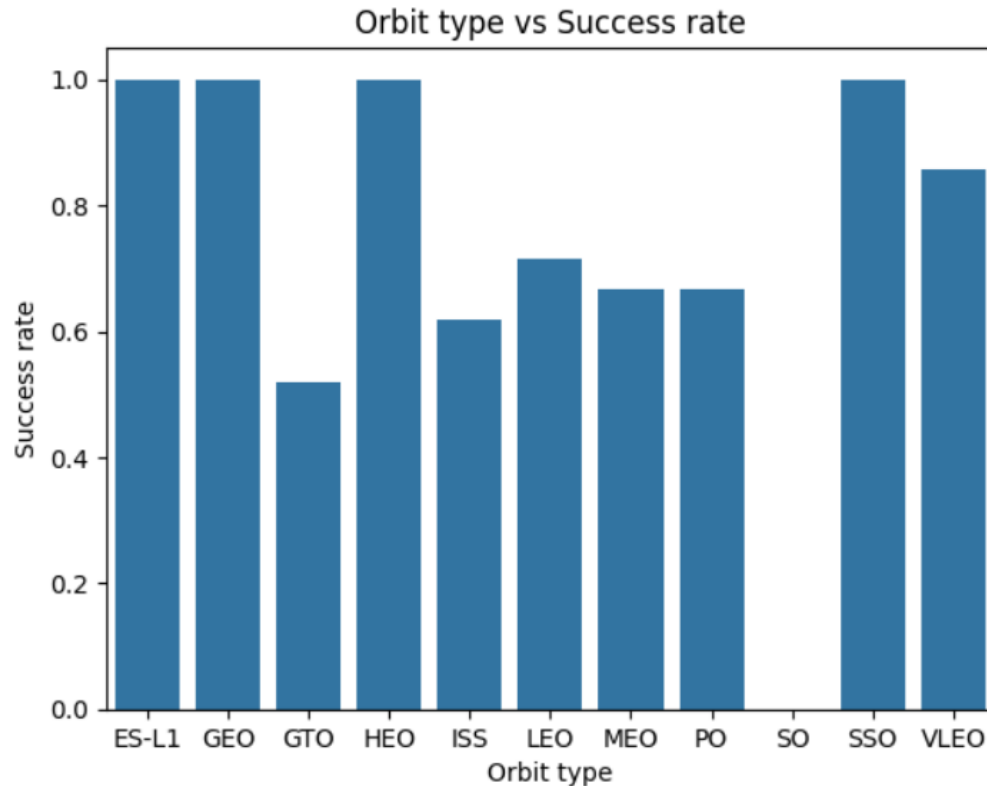
Key Insights:

- **Overall Trend:** the higher payload mass tended to have a greater success rate for every launch site.
- **CCAFS SLC 40:** The success rate for payload mass between 0 and 7,500 kg was about 50%, while for higher payload the success rate increased drastically.
- **VAFB SLC 4E:** It had the highest success rate with only two failed attempts across every payload mass. Interestingly, the maximum payload mass for this launch site was 10,000 kg.
- **KSC LC 39A:** This landing site also had a high success rate with 3 failures between 5,000 and 7,500 kg and only one failed attempt for payload mass more than 15,000 kg.

Figure 2: Payload Mass (kg) vs Launch Site landing outcome distribution.

- URL: <https://github.com/jingsann/Falcon-9-project/blob/8db94669c76026d1c6d05efbe03acf18b2aa1886/Spacex%20data%20preparation.ipynb>

EDA with Data Visualisation



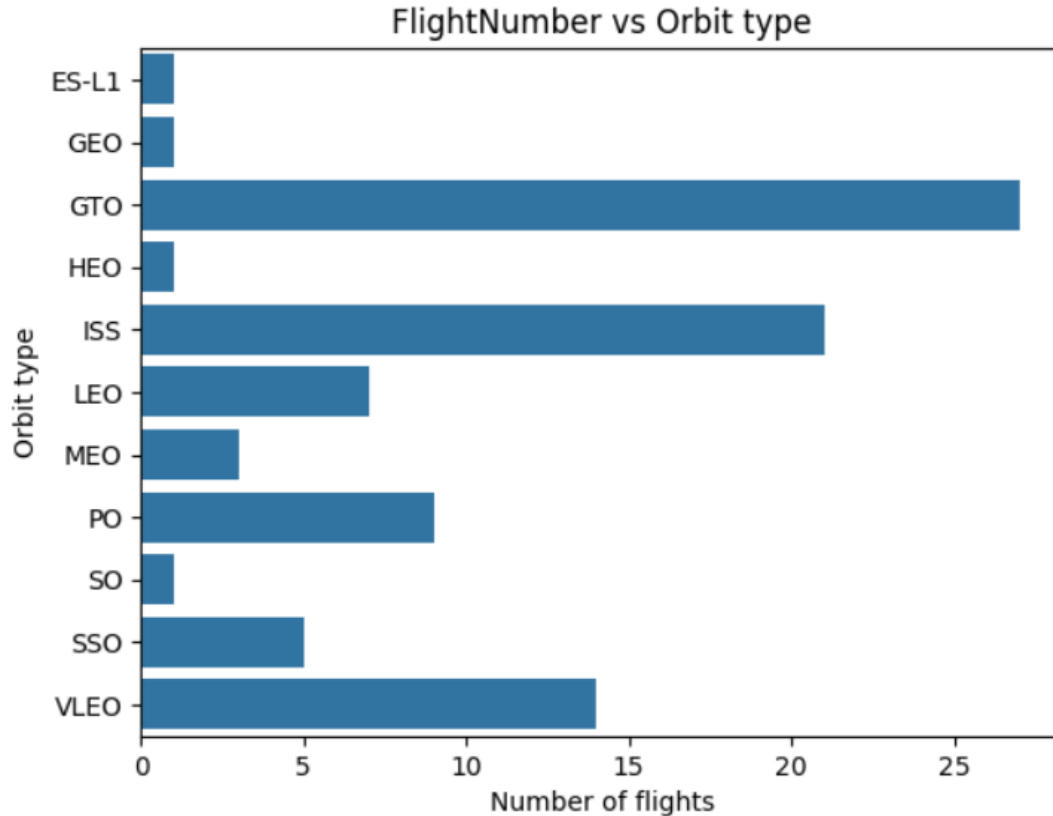
Key Insights:

- Overall, the success rate for every orbit was higher than 50% except SO, which was not successful at all.
- Launches to ES-L1, GEO, HEO, and SSO orbits were completely successful, followed by VLEO orbit at about 80%.
- For LEO, MEO, PO, and ISS orbits, the success rate was approximately 60% on average.
- Launches to GTO orbit recorded the second-lowest success rate at around 50%.

Figure 3: The relationship between success rate of each orbit type.

- URL:<https://github.com/jingsann/Falcon-9-project/blob/8db94669c76026d1c6d05efbe03acf18b2aa1886/Spacex%20data%20preparation.ipynb>

EDA with Data Visualisation



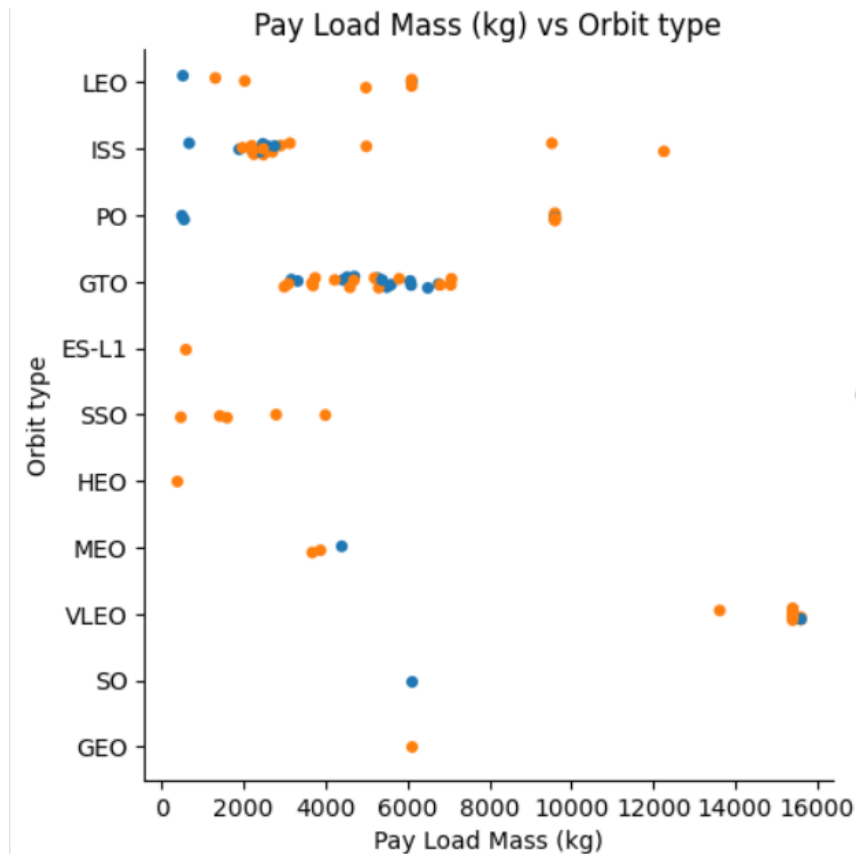
Key Insights:

- **Overall Distribution:** GTO, ISS, and VLEO recorded the highest flight counts, collectively accounting for more than half of all missions.
- **Moderate Frequency:** PO, LEO, and SSO orbits saw moderate activity, averaging approximately 7 flights each.
- **Minimal Activity:** ES-L1, GEO, HEO, and SO orbits had only one mission each.

Figure 4: The relationship between FlightNumber and Orbit type.

- URL:<https://github.com/jingsann/Falcon-9-project/blob/8db94669c76026d1c6d05efbe03acf18b2aa1886/Spacex%20data%20preparation.ipynb>

EDA with Data Visualisation



Key Insights:

- **Overall Distribution:** VLEO missions recorded the highest average payload, whereas ES-L1 and HEO carried the lowest weights.
- **Payload Consistency:** Launches to SO and GEO shared an identical payload mass of approximately 6,000 kg.
- **Moderate Cargo Ranges:** GTO, PO, ISS, and LEO flights averaged around 4,500 kg, whilst MEO missions typically carried about 4,000 kg.

Figure 5: The relationship between Payload Mass and Orbit type.

- URL: <https://github.com/jingsann/Falcon-9-project/blob/8db94669c76026d1c6d05efbe03acf18b2aa1886/Spacex%20data%20preparation.ipynb>

EDA with Data Visualisation

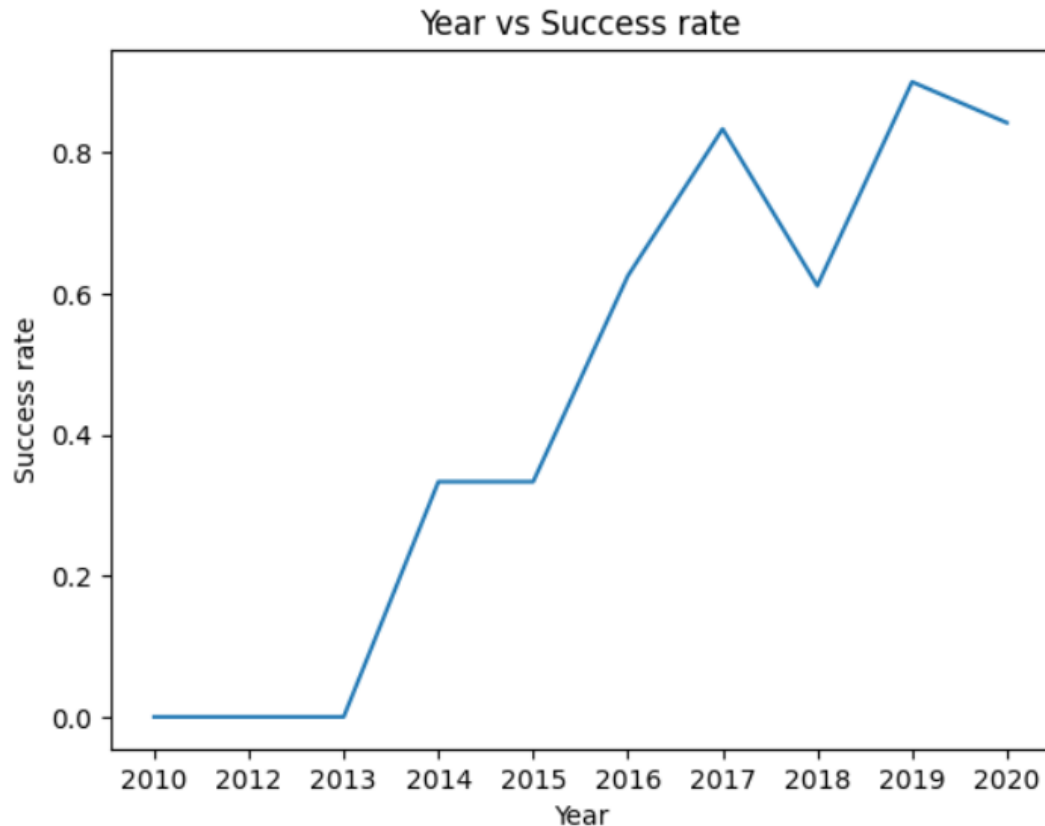


Figure 6: The launch success yearly trend.

Key Insights:

- **Overall Trend:** The success rate improved progressively over time. There were 3 development stages.
- **Initial Phase (2010-2013):** There were no successful landings.
- **Rapid Growth Phase (2013-2017):** There was a dramatic rise in the success rate up to 80%.
- **Mature Phase (2017-2020):** The success rate increased slightly with a dip to about 60% in 2018.

- URL: <https://github.com/jingsann/Falcon-9-project/blob/8db94669c76026d1c6d05efbe03acf18b2aa1886/Spacex%20data%20preparation.ipynb>

EDA with SQL

Key Takeaways:

- There were 4 unique launch sites: CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, and CCAFS SLC-40.
- The total payload mass carried by boosters launched by NASA (CRS) was 45,596 kg.
- The average payload mass carried by booster version F9 v1.1 was 2,928.4 kg.
- The first successful ground pad landing was achieved on 22 December 2015.
- There were 14 different boosters that landed successfully on a drone ship with payload masses between 4,000 and 6,000 kg.
- URL: https://github.com/jingsann/Falcon-9-project/blob/4a37e516252e814ca6d9be837e05791acfe55423/jupyter-labs-eda-sql-coursera_sqllite.ipynb

EDA with SQL (continued)

Key Takeaways:

- Recorded 100 successful launch outcomes, with one flight having an unknown payload mass and one recorded failure.
- Booster F9 B5 B1048.4 carried the highest payload mass at 15,600 kg.
- Two consecutive drone-ship landing failures occurred at CCAFS LC-40 in 2015, involving boosters F9 v1.1 B1012 in January and F9 v1.1 B1015 in April.
- Between 4 June 2010 and 20 March 2017, the fleet recorded 38 general successful outcomes, 14 drone-ship landings, 9 ground-pad landings, 22 flights with no landing attempt, and 18 failed or uncontrolled outcomes.
- URL: https://github.com/jingsann/Falcon-9-project/blob/4a37e516252e814ca6d9be837e05791acfe55423/jupyter-labs-eda-sql-coursera_sqlite.ipynb

Folium Visualisation

Key Takeaways:

- CCAFS LC-40, KSC LC-39A, and CCAFS SLC-40 are located in close proximity to one another in Florida, whereas VAFB SLC-4E is situated in California.
- The majority of launches took place in Florida, comprising 33 missions from CCAFS LC-40 and CCAFS SLC-40, and 13 from KSC LC-39A. In contrast, only 10 launches occurred in California.
- CCAFS LC-40 and CCAFS SLC-40 are situated 590 m from the nearest highway, 900 m from the coastline, 970 m from the railway, and 55.23 km from the closest city (Melbourne, Florida). These distances provide a safe buffer zone for launch operations.
- URL: <https://github.com/jingsann/Falcon-9-project/blob/4a37e516252e814ca6d9be837e05791acfe55423/Visualisation%20with%20Folium.ipynb>

Plotly Dash (Pie Chart Section)

Key Takeaways:

- Across all sites, the most successful launch site was KSC LC-39A, accounting for 40%. This was followed by CCAFS LC-40 at nearly 30% and VAFB SLC-4E at about 15%, while the lowest was CCAFS SLC-40 at about one-eighth.
 - Approximately three-quarters of launches at the CCAFS LC-40 and KSC LC-39A launch sites were successful.
 - Exactly 60% of launches at the VAFB SLC-4E launch site were successful. CCAFS SLC-40 also experienced a similar trend.
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- URL: <https://github.com/jingsann/Falcon-9-project/blob/4a37e516252e814ca6d9be837e05791acfe55423/spacex-dash-app.py>

Plotly Dash (Scatter Plot Section)

Key Takeaways:

- Booster v1.0 performance was the lowest because there were no successful launches, and the maximum payload mass was 700 kg.
- An upgrade to booster v1.1 increased the threshold of payload mass to 4,500 kg; however, a launch with a 2,000 kg payload was the only successful attempt out of 15.
- Booster FT performance was inconsistent across all payload masses; in other words, exactly half of all attempts were successful.
- Booster B4 demonstrated strong performance for payload masses around 4,000 kg.
- The performance of booster B5 remains inconclusive because there was only one successful attempt.
- URL: <https://github.com/jingsann/Falcon-9-project/blob/4a37e516252e814ca6d9be837e05791acfe55423/spacex-dash-app.py>

Predictive analysis by different models

ML technique	Accuracy	R2 test set
Logistic Reg	0.85	0.83333
SVMs	-	-
Decision Tree	0.892	0.72222
Knn neighbour	0.64286	0.72222

Key Takeaways:

- Support Vector Machines (SVMs) took too long to train and failed to yield results; therefore, they performed the worst among the evaluated methods.
- Logistic Regression is the best-performing model due to its optimal balance of a high R2 score (0.83) and strong accuracy (0.85), followed closely by Decision Tree (which had higher raw accuracy but a lower R2) and K-Nearest Neighbours (KNN).

- URL: https://github.com/jingsann/Falcon-9-project/blob/4a37e516252e814ca6d9be837e05791acfe55423/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb